

ARDUINO INTERFACING UTILITIES

University of Alexandria, Faculty of Engineering,
Computer and Systems Engineering Department.
CSE343 Embedded Systems Spring 2026

Tone Function

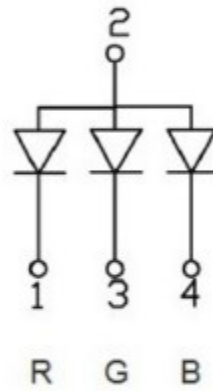
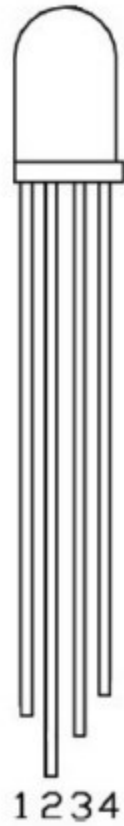
- Description
 - Generates a square wave of the specified frequency (and 50% duty cycle) on a pin. A duration can be specified, otherwise the wave continues until a call to noTone(). The pin can be connected to a piezo buzzer or other speaker to play tones.
 - Tone is used with PWM output pins.
- Syntax
 - `tone(pin, frequency)`
 - `tone(pin, frequency, duration)`
- Parameters
 - `pin`: the Arduino pin on which to generate the tone.
 - `frequency`: the frequency of the tone in hertz. Allowed data types: unsigned int.
 - `duration`: the duration of the tone in milliseconds (optional). Allowed data types: unsigned long.

Interfacing a Buzzer

```
const int buzzerPin = 9;
void setup(){
  pinMode(buzzerPin, OUTPUT);
}

void loop(){
  tone(buzzerPin, 1500, 500);
  delay(1000);
}
```

RGB LED



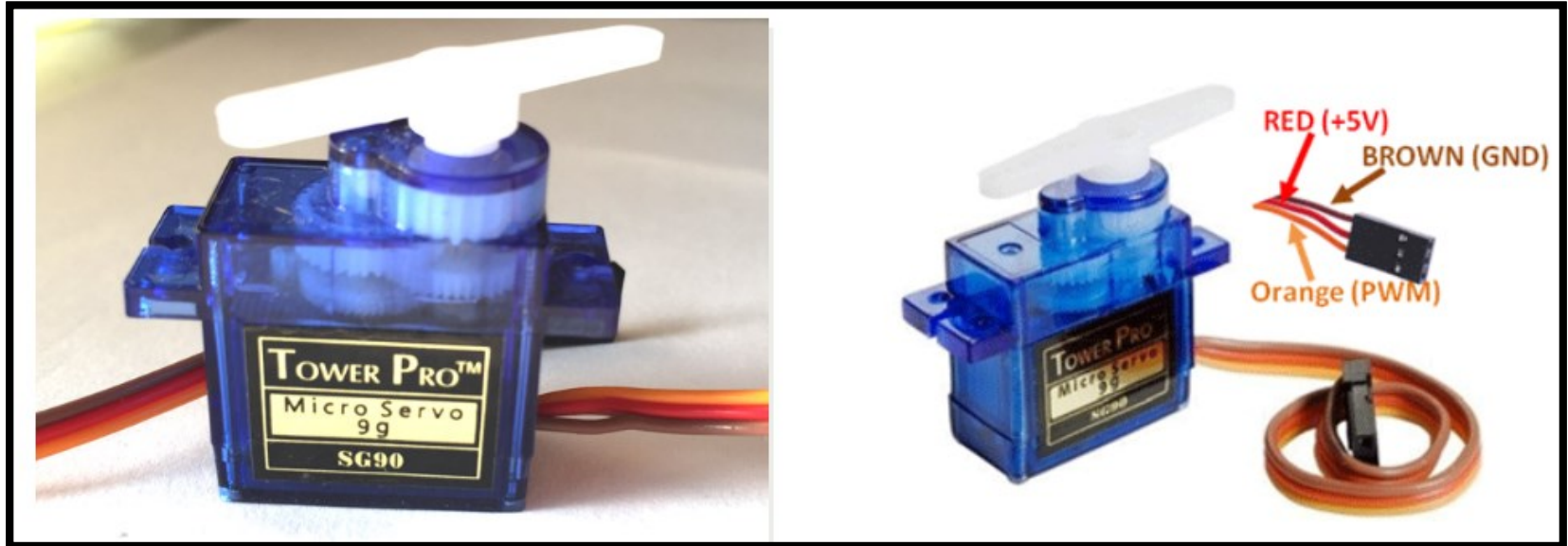
Interfacing RGB LED

```
const int redPin = 2;
const int greenPin = 3;
const int bluePin = 4;

void setup() {
    pinMode(redPin, OUTPUT);
    pinMode(greenPin, OUTPUT);
    pinMode(bluePin, OUTPUT);
}

void loop() {
    // put your main code here, to run repeatedly:
    digitalWrite(redPin, LOW);
    digitalWrite(greenPin, HIGH);
    digitalWrite(bluePin, HIGH);
    delay(1000);
    digitalWrite(redPin, HIGH);
    digitalWrite(greenPin, LOW);
    digitalWrite(bluePin, HIGH);
    delay(1000);
    digitalWrite(redPin, HIGH);
    digitalWrite(greenPin, HIGH);
    digitalWrite(bluePin, LOW);
    delay(1000);
    digitalWrite(redPin, LOW);
    digitalWrite(greenPin, LOW);
    digitalWrite(bluePin, LOW);
    delay(1000);
}
```

Servomotor



Interfacing Servomotor

```
//Written By DKV 3-8-19, Servo tester//
#include <Servo.h>
Servo servo1; //first servo attached to A0

int servoPos1 = 0; //starting positions for servos

void setup() {
  servo1.attach(A0); //servo attaching pins (A0)
};

void loop() {
  //SERVO 1
  for(servoPos1 = 0; servoPos1 < 180; servoPos1++) //When servo is a position 0, servo will rotate to 180 degrees (half rotation)
  {
    servo1.write(servoPos1);
    delay(10);
  };

  for(servoPos1 = 180; servoPos1 > 0; servoPos1--) //When servo is a position 180, servo will rotate to 0 degrees (beginning position)
  {
    servo1.write(servoPos1);
    delay(10);
  };
};
```

References

- **Tone Function**

- [tone\(\) - Arduino Reference](#)

- **Common anode RGB specifications:**

- [LED5D-RGB-CA.pdf - Google Drive](#)

- **Servomotor**

- [Servo Motor SG-90 Basics, Pinout, Wire Description, Datasheet, and Working \(components101.com\)](#)